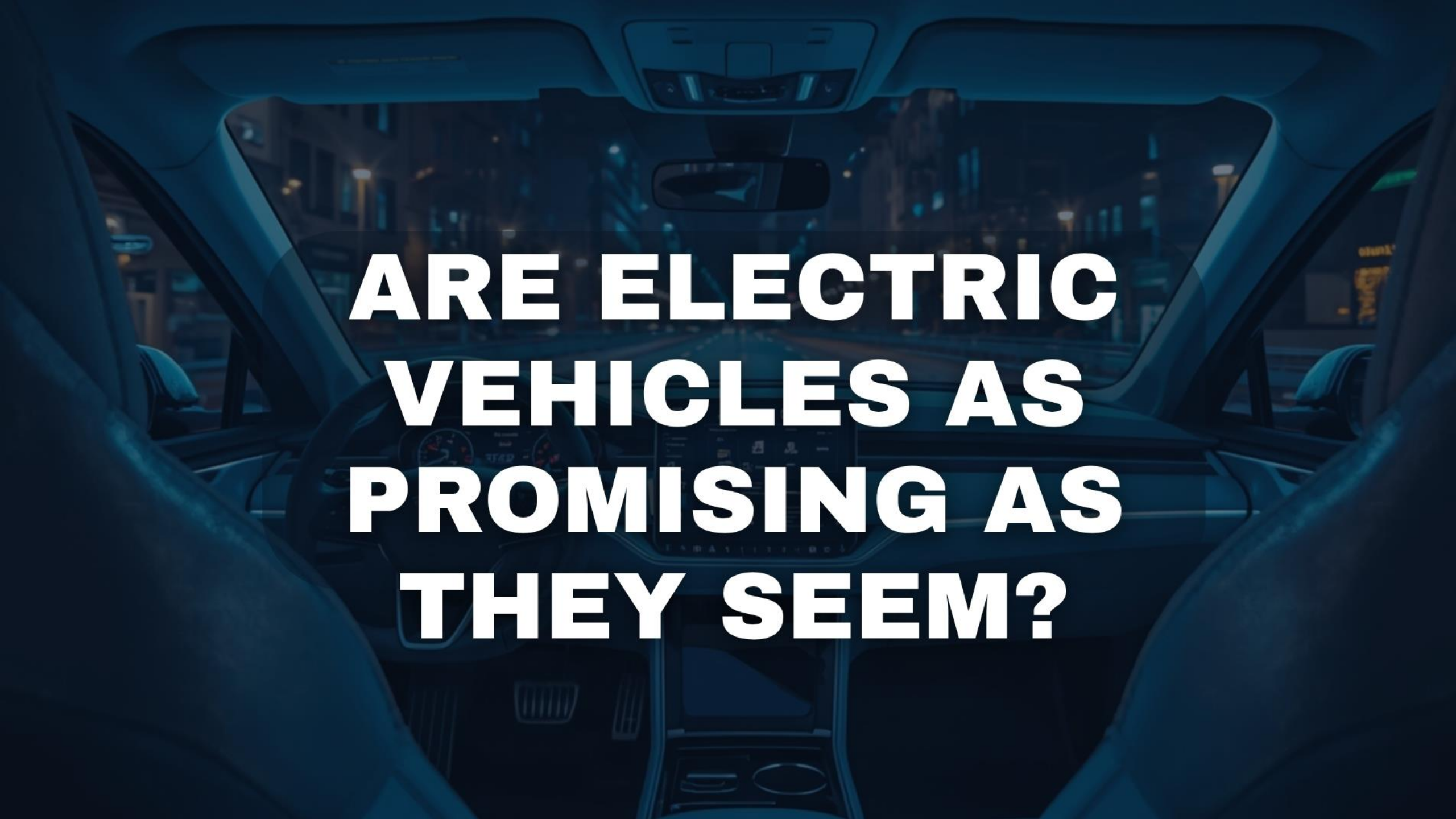
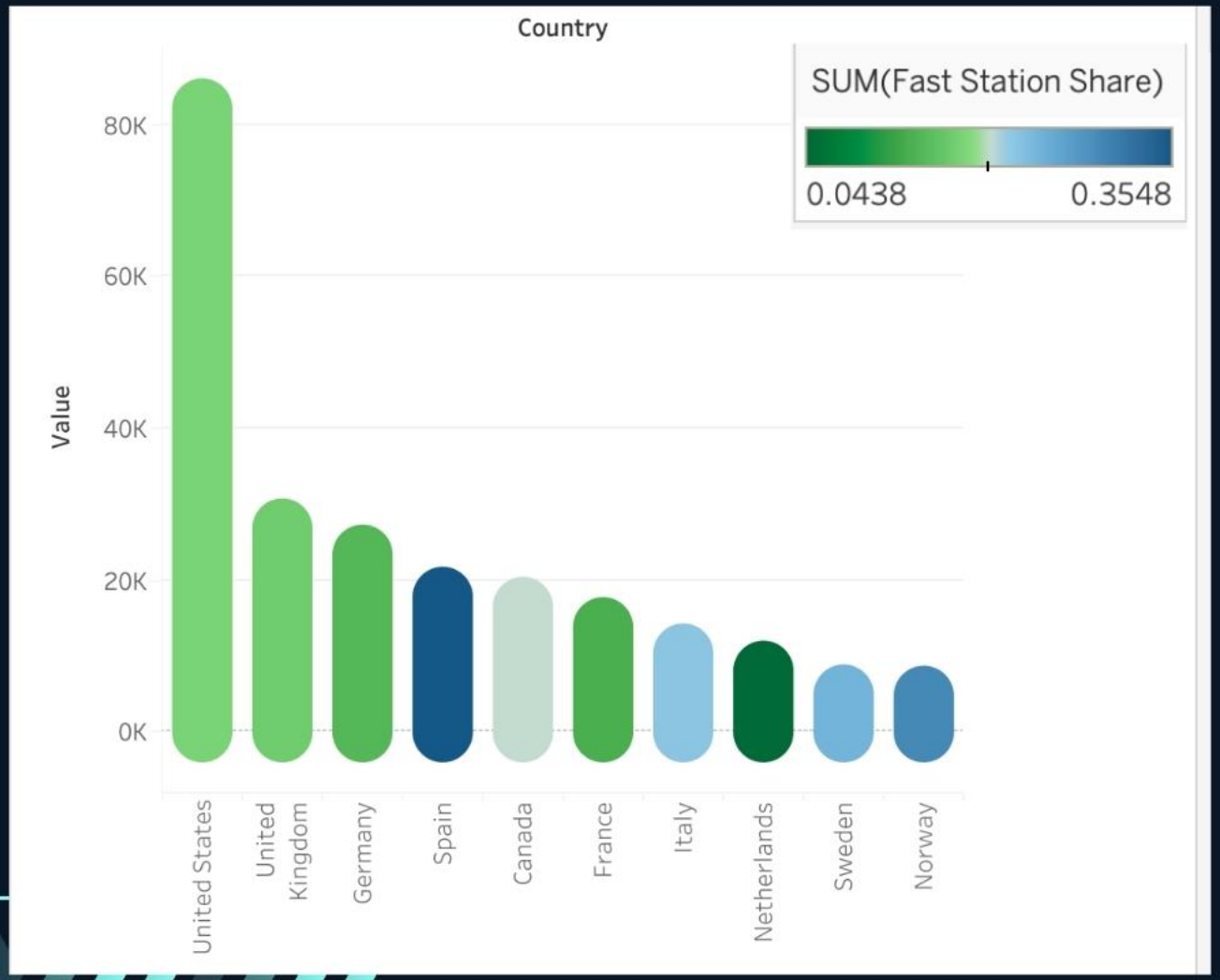
THE FUTURE OF THE EV



DATALICIOUS TEAM

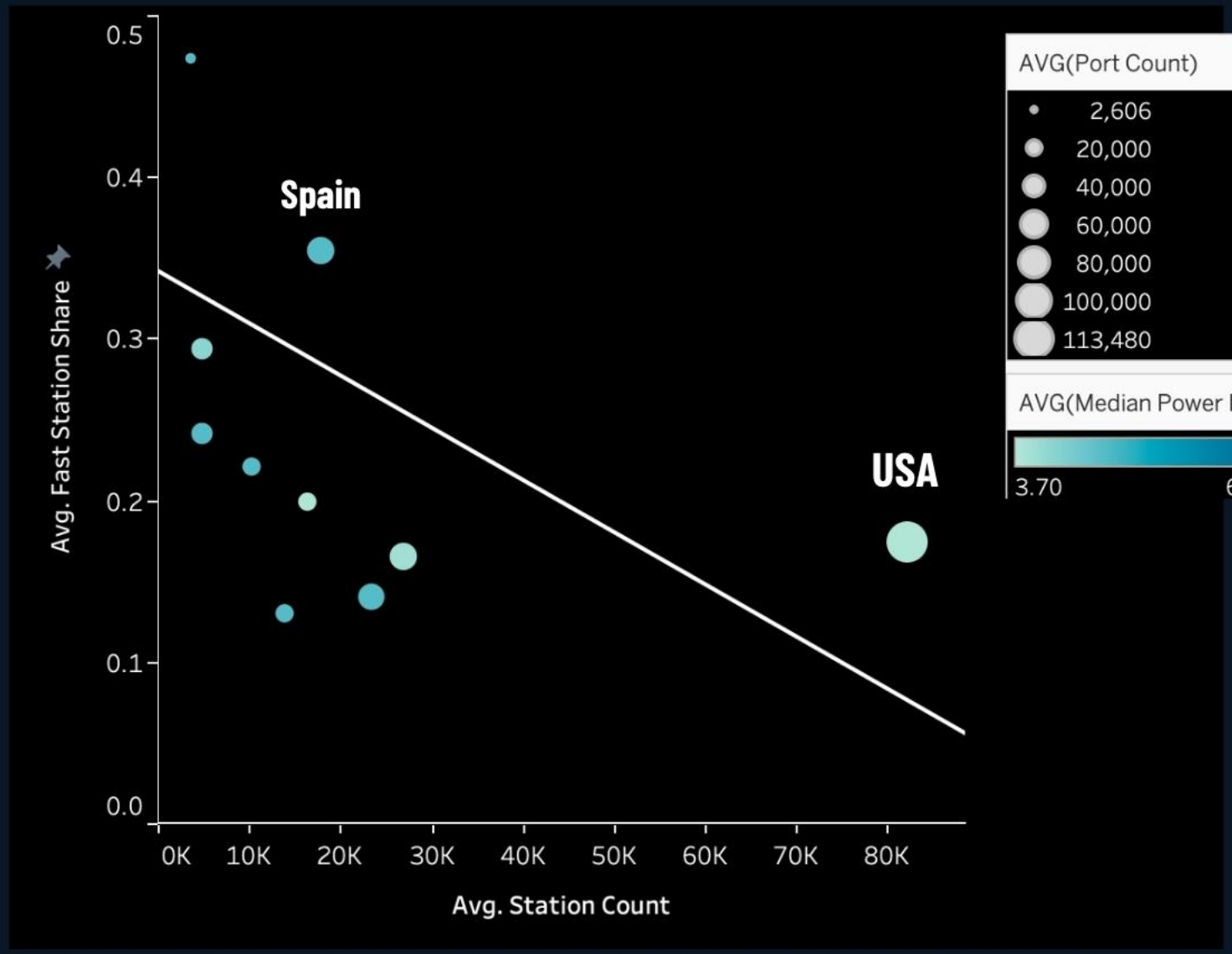

ARE ELECTRIC VEHICLES AS PROMISING AS THEY SEEM?

TOP 10 COUNTRIES BY AVERAGE CHARGING STATIONS





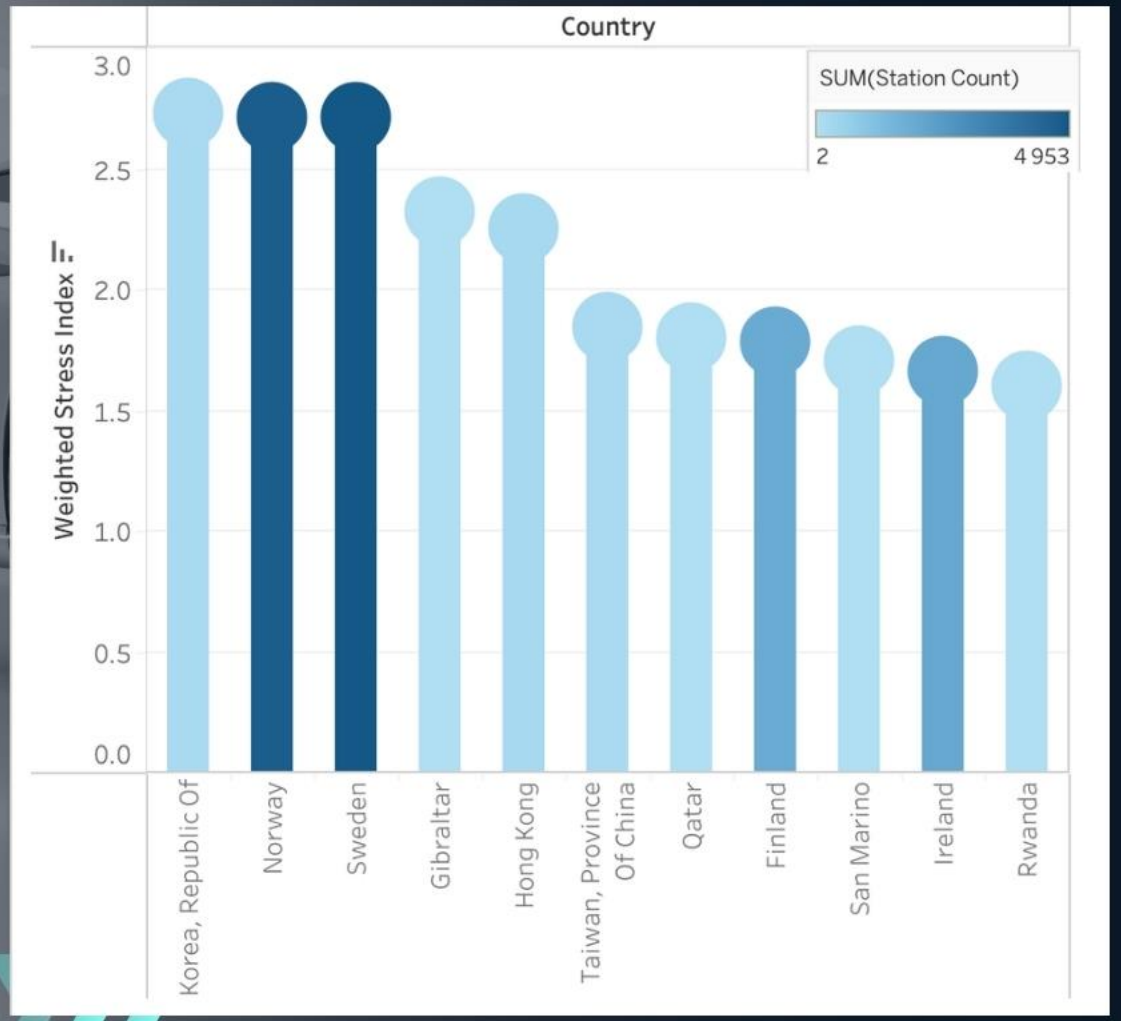
FAST STATION SHARE VS STATION COUNT



The more stations they country has, the lower its fast station share



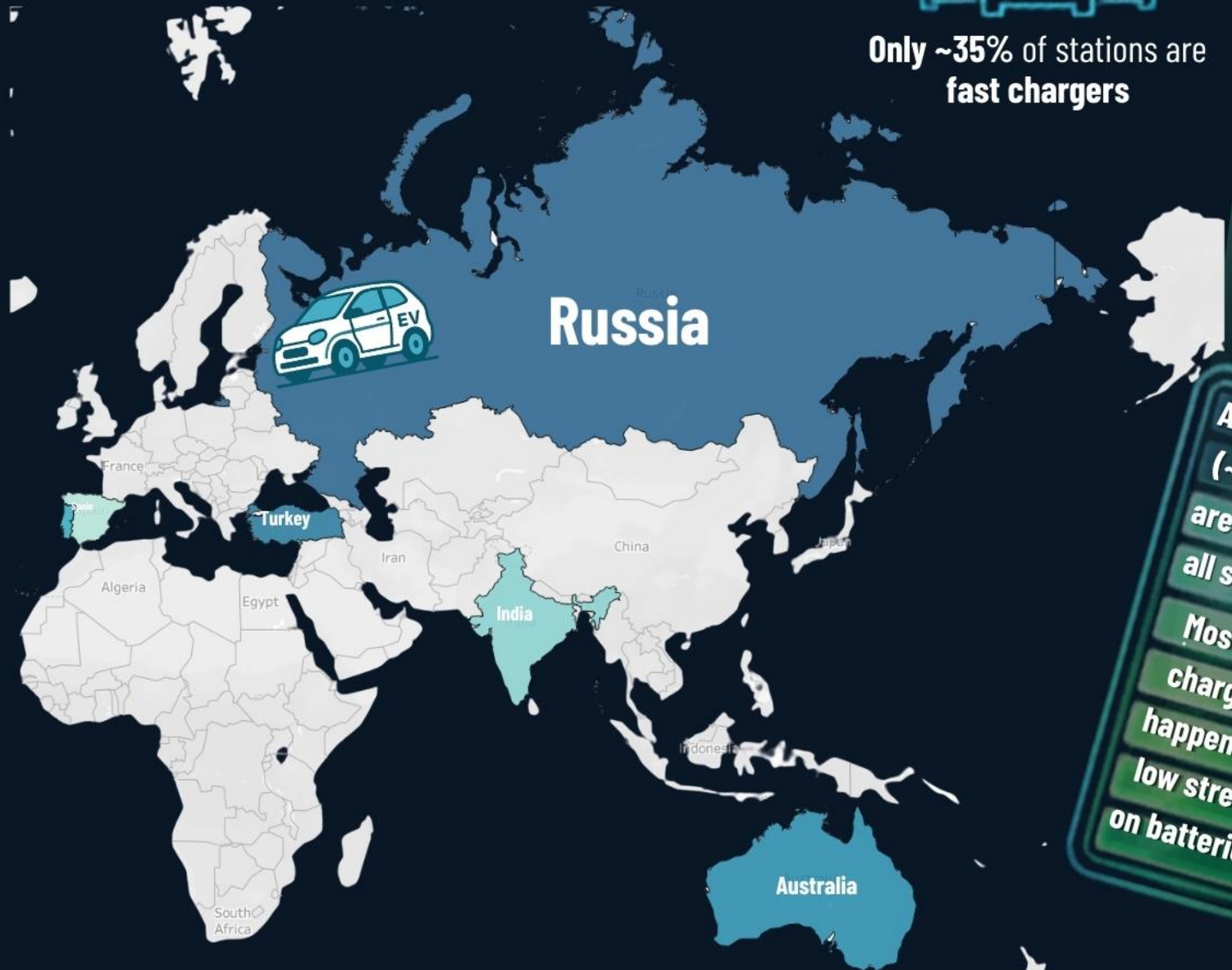
WHICH COUNTRIES FACE THE MOST INFRASTRUCTURE STRESS?



FAST CHARGING LEADERS WORLDWIDE



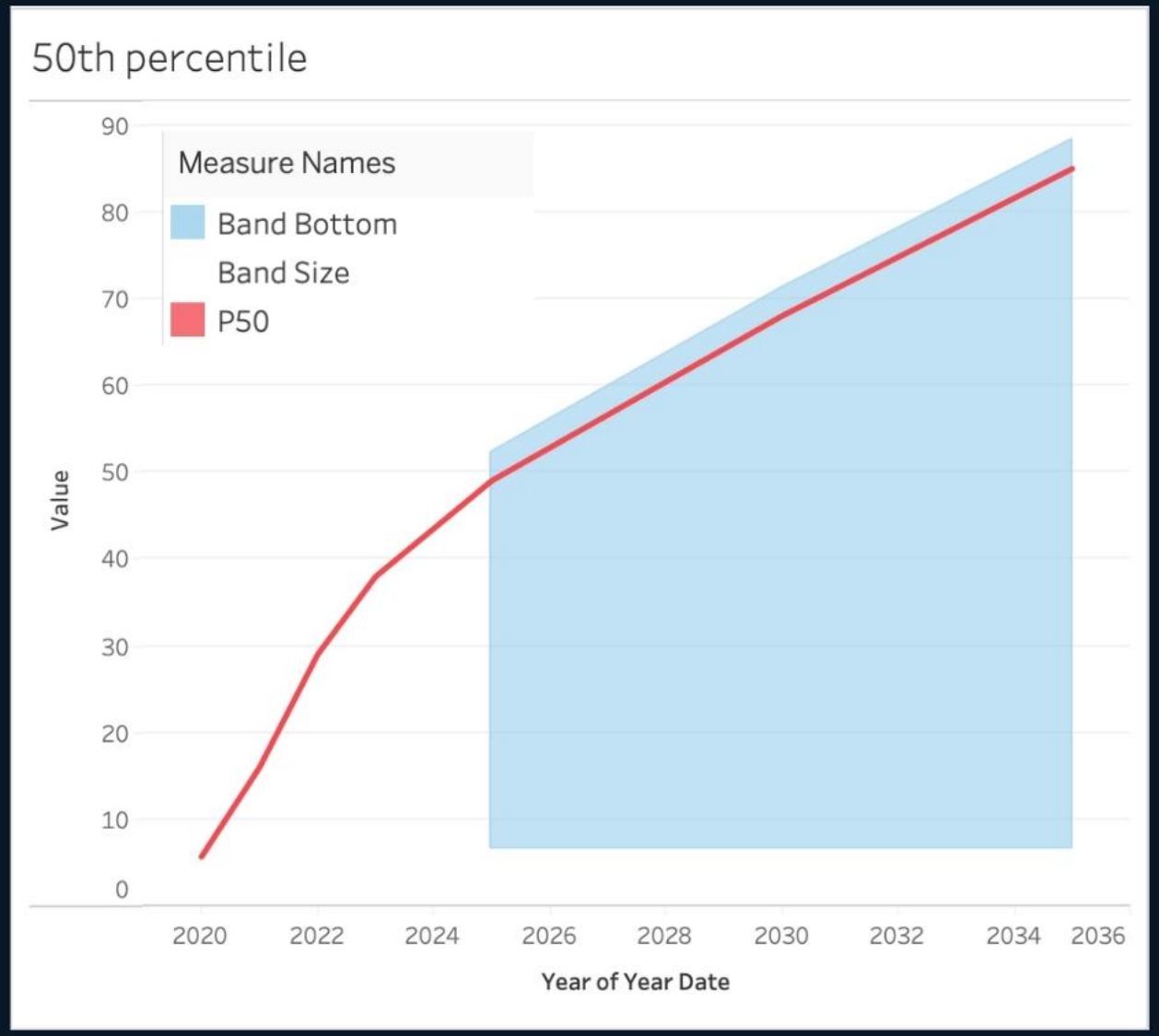
Only ~35% of stations are **fast chargers**



AC chargers (~7-49kW) are +75% of all stations

Most EV charging happens at low stress on batteries

EV ADOPTION FORECAST USING UNCERTAINTY BAND



EV Sales jump from **~10,000** in 2010 to **143 mil.** by 2035

Significant sales increase



```r
REGIONS <- list(
  World = list(
    hist  = c(`2020` = 4.2, `2021` = 8.9, `2022` = 14, `2023` = 18),
    aps   = c(`2025` = 26, `2030` = 48, `2035` = 68),
    steps = c(`2025` = 25, `2030` = 44, `2035` = 55)
  ),
  China = list(
    hist  = c(`2020` = 5.7, `2021` = 16, `2022` = 29, `2023` = 38),
    aps   = c(`2025` = 49, `2030` = 68, `2035` = 85),
    steps = c(`2025` = 49, `2030` = 68, `2035` = 85)
  ),
  Europe = list(
    hist  = c(`2020` = 10, `2021` = 17, `2022` = 20, `2023` = 21),
    aps   = c(`2025` = 31, `2030` = 62, `2035` = 92),
    steps = c(`2025` = 31, `2030` = 60, `2035` = 87)
  ),
  USA = list(
    hist  = c(`2020` = 2.3, `2021` = 4.7, `2022` = 7.4, `2023` = 9.5),
    aps   = c(`2025` = 20, `2030` = 55, `2035` = 71),
    steps = c(`2025` = 20, `2030` = 55, `2035` = 71)
  )
)
set.seed(42)
N <- 5000
rows <- list()
for (region in names(REGIONS)) {
  d <- REGIONS[[region]]
  # Historical rows — no uncertainty
  for (year in names(d$hist)) {
    val <- d$hist[[year]]
    rows[[length(rows) + 1]] <- data.frame(
      Region = region,
      Year = as.integer(year),
      Type = "Historical",
      P5 = val,
      P50 = val,
      P95 = val,
      APS = val,
      STEPS = val
    )
  }
  # Projection rows — simulate
  proj_years <- sort(as.integer(names(d$aps)))
  for (year in proj_years) {
    year_chr <- as.character(year)
    mu <- (d$aps[[year_chr]] + d$steps[[year_chr]]) / 2
    gap <- abs(d$aps[[year_chr]] - d$steps[[year_chr]])
    sigma <- max(gap / 2, 2)  # volatility scales with scenario spread

    samples <- rnorm(N, mean = mu, sd = sigma)
    samples <- pmin(pmax(samples, 0), 100)

    rows[[length(rows) + 1]] <- data.frame(
      Region = region,
      Year = year,
      Type = "Projection",
      P5 = as.numeric(quantile(samples, 0.05)),
      P50 = as.numeric(quantile(samples, 0.50)),
      P95 = as.numeric(quantile(samples, 0.95)),
      APS = d$aps[[year_chr]],
      STEPS = d$steps[[year_chr]]
    )
  }
}
df <- do.call(rbind, rows)
write.csv(df, "ev_mc_percentiles.csv", row.names = FALSE)
```


SO, WHAT'S THE TRUTH?

WE ARRIVED AT OUR FINAL DESTINANTION!

